Do Now · retrieval
Stacks and queues

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a[1] for int[] a={6,9,3}?

Q2

What is −1 in 8-bit two's complement?

Q3

A loop running from 0 to 3 inclusive executes how many times?

Q4

Push 2, push 5, pop once: which value is removed?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Push — Adding an item puts it on top of a stack.

Core idea 2

Pop — Removing an item takes the most recently pushed item: 7.

Core idea 3

Contrast with a queue — A queue removes the oldest item instead (FIFO); it is suited to arrival-order processing.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

stack — LIFO abstract data type.

push — Add an item onto a stack.

pop — Remove the top stack item.

queue — FIFO abstract data type.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

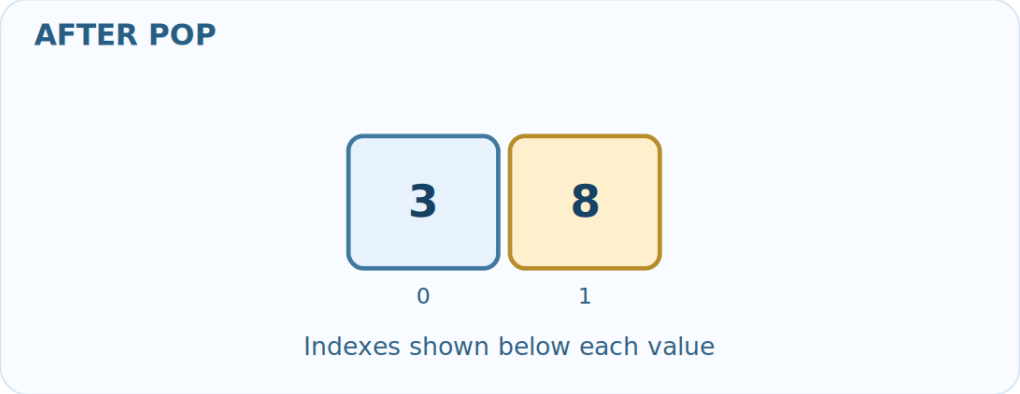
Problem

A stack receives 3, 8, 5, then Pop once. Give removed value and remaining stack bottom-to-top.

Follow the worked method

1. Predict — The newest value 5 is on top, so Pop removes 5.
2. Confirm and explain — The stack remains [3,8] from bottom to top.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

After A,B,C: front=0, rear=0, count=3. The queue is full even though front==rear. rear means next free slot here, unlike Lesson 020.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Push A, B, C onto a stack then Pop twice. What remains?

Question 2

A circular array queue has capacity 3. Starting empty, enqueue 10,20,30; dequeue once; enqueue 40. State physical slots, front/rear/count and logical order. Use rear as the next free slot.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why an undo history usually uses a stack rather than a queue.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A queue removes the most recently inserted item first.

Use key vocabulary from memory

1. Explain stack in your own words.

2. Explain push in your own words.

3. Explain pop in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.